

Identification of Spatial Patterns in Wildfire Hotspots in Brazil: A Comparative Approach Among K-Means, GMM, and DBSCAN for SDG 15

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Abstract—The monitoring and mitigation of wildfires are central challenges for biodiversity preservation, aligned with the Sustainable Development Goals (SDG 15 – Life on Land). This paper proposes an unsupervised spatial analysis on a real-world dataset from INPE (BDQueimadas) containing over 284 thousand fire hotspots in Brazil, of which 136 thousand with valid FRP ($n > 1000$). Unlike purely geometric approaches, the feature vector was extended with the Fire Radiative Power (FRP), enabling algorithms to identify spatial clusters of high destructive intensity. Three clustering algorithms were implemented and compared: K-Means, *Gaussian Mixture Model* (GMM), and DBSCAN. The statistical evaluation, conducted over 30 independent runs, demonstrated the methodological superiority of DBSCAN, which achieved a mean Silhouette Coefficient of 0.4619 compared to 0.3554 for K-Means and 0.0474 for GMM. A Kruskal-Wallis hypothesis test ($p < 10^{-12}$) confirmed that the difference is statistically significant. The results show that the non-linear and asymmetric nature of fire propagation requires density-guided models, overcoming the topological limitations of spherical or elliptical clusters.

Index Terms—Artificial Intelligence, Data Mining, DBSCAN, K-Means, SDG 15, Wildfires, Silhouette Coefficient.

I. INTRODUCTION

The progressive collapse of terrestrial biomes, catalyzed by the spread of uncontrolled wildfires, threatens endemic biodiversity and exacerbates greenhouse gas emissions [8]. Containing these catastrophes depends substantially on the analytical capacity to predict and detect spatiotemporal degradation patterns, which places artificial intelligence (AI) as an indispensable tool for achieving the 2030 Agenda goals, specifically SDG 15.

In this context, the present work aims to apply spatial data mining techniques [1], [9] to identify endemic wildfire zones in Brazilian territory. The research goes beyond elementary algorithm application by using real microdata with a large number of instances.

The central contribution of this study lies in the empirical comparative evaluation between parametric models that assume predefined cluster shapes (such as K-Means and GMM) and density-based models (DBSCAN), which map the organic continuity of affected areas [2]. Additionally, we expand the classical scope by incorporating Fire Radiative Power (FRP) into the feature vector and conducting a segmented analysis by biome, contrasting fire morphology in the Cerrado and the

Amazon. The paper is organized as follows: Section II discusses the relevant literature; Section III details data engineering, mathematical foundations, and the algorithms employed; Section IV presents the experiments and statistical results; and, finally, Section V closes with conclusions and future work.

II. RELATED WORK

Contemporary literature presents several computational approaches focused on monitoring forest degradation. To situate this work, five studies of theoretical and applied relevance are highlighted.

First, in the field of partitional clustering, authors applied unsupervised K-Means on Sentinel satellite imagery to detect deforestation polygons in the Amazon [5]. The problem addressed was the automatic segmentation of degraded areas from spectral bands, and the authors obtained reasonable discretization of the regions. However, the method is hampered by its susceptibility to noise and its assumption of convex clusters, artificially grouping disconnected areas solely by linear proximity. Our approach uses K-Means only as a baseline and empirically demonstrates this limitation against the organic shapes of fire.

Second, the DBSCAN algorithm was originally proposed as a density-based clustering method for discovering classes in large spatial databases with noise [2]. The central problem addressed by the authors was identifying clusters of arbitrary shape without requiring prior definition of the number of groups and with minimal domain knowledge requirements, treating isolated points as noise. We build on this theoretical foundation by transposing it to the wildfire domain, extending the feature vector with FRP to search not only for spatial clusters but for concentrations of high radiative intensity, and validating clusters via the Silhouette Coefficient and hypothesis testing.

Third, advancing in the temporal dimension, the ST-DBSCAN (*Spatial-Temporal DBSCAN*) algorithm extended DBSCAN through three modifications that allow grouping objects considering non-spatial, spatial, and temporal values simultaneously [7]. This formulation processes data not as static snapshots, but incorporating temporal continuity, making it promising for mapping propagation corridors. Our proposal is deliberately spatial and intensity-based (lower computational

cost), leaving the temporal component as future work supported by this algorithm.

Fourth, in the structured predictive field, a supervised and department-sensitive approach was proposed for predicting wildfire risk, arguing that risk should not be assumed uniform across regions, but adjusted to the local conditions of each territorial unit [8]. The authors introduced a new dataset and the first national AI benchmark at metropolitan scale for France, employing state-of-the-art models. In contrast, our method is descriptive and unsupervised, requiring no risk labels and offering lower annotation cost pattern exploration; nevertheless, we converge with that work in finding, in our biome analysis, that regionalized model calibration is decisive.

Finally, the use of Deep Learning was investigated for detecting forest anomalies in the Amazon and Cerrado biomes [6]. Convolutional Neural Networks (CNNs) were used to segment hotspots with high precision, at the cost of overwhelming computational power and large labeled repositories. In contrast, our solution operates only on vector coordinates (Latitude and Longitude) augmented with FRP, offering drastically lower computational cost, suitable for environmental security agencies with limited resources.

III. METHODOLOGY

The methodological foundation operated at the convergence of environmental preprocessing and unsupervised clustering parameterization.

A. Spatial Preprocessing

The raw matrix underwent strict removal of corrupted data (missing values encoded as -999). The feature vector was composed of three dimensions: Latitude, Longitude, and FRP. To prevent distance-sensitive algorithms from having their performance degraded by scale disparity (geographic coordinates *versus* radiative power in MW), Z-score standardization (*StandardScaler*) was compulsorily applied, so that each axis was given zero mean and unit variance [1].

B. Mathematical Foundations

K-Means. A classic partitional method [1], [9], the algorithm partitions the dataset into k groups by minimizing inertia, i.e., the sum of squared Euclidean distances from each point to the centroid μ_j of its cluster C_j :

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2. \quad (1)$$

The iterative minimization of (1) assumes convex, homogeneous-density clusters, making it sensitive to *outliers* and elongated shapes.

DBSCAN. Density-based clustering [2] requires no prior definition of the number of groups. A point p is classified as a *core* point if its ϵ -neighborhood contains at least $MinPts$ instances:

$$|N_\epsilon(p)| \geq MinPts, \quad N_\epsilon(p) = \{q \mid d(p, q) \leq \epsilon\}. \quad (2)$$

Clusters are expanded by density-reachability from core points, while unreachable instances are labeled as noise. This mechanism allows tracking arbitrary fire corridors and isolating scattered hotspots.

Silhouette Coefficient. The intrinsic quality of clusters was measured by the Silhouette Coefficient [3]. For each instance i , let $a(i)$ be the mean intra-cluster distance and $b(i)$ the smallest mean distance to a neighboring cluster. The silhouette of i is:

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad S(i) \in [-1, 1]. \quad (3)$$

The global index is the mean of $S(i)$ over all instances; values close to 1 indicate cohesive and well-separated clusters.

C. Decision Mechanisms and Algorithms

- **K-Means:** partitional algorithm that defines spherical clusters by minimizing the inertia of Eq. (1) (empirically set to $k = 8$).
- **Gaussian Mixture Model (GMM):** probabilistic extension of K-Means that estimates multivariate Gaussian distributions. The covariance matrix was set to full, allowing elliptical detection shapes.
- **DBSCAN:** density-based clustering with neighborhood radius $\epsilon = 0.30$ and minimum cardinality $MinPts = 15$. These hyperparameters were selected by systematic grid search over candidate values in the three-dimensional space (Lat/Long/FRP), adopting the configuration that maximizes the Silhouette Coefficient, as described in the parameter variation study presented in Section IV.

IV. EXPERIMENTAL ANALYSIS

A. Dataset

The research used the consolidated governmental record (BDQueimadas) from the National Institute for Space Research (INPE), covering the year 2025 [4]. The dataset contains 284,152 geolocated hotspots across the Amazon, Cerrado, Caatinga, Atlantic Forest, Pantanal, and Pampa biomes. After requiring valid FRP for the composition of the three-dimensional feature vector, 136,393 instances composed the analysis, far exceeding the course requirement ($n > 1000$).

The exploratory analysis (Fig. 1) reveals two determining characteristics. The FRP distribution is strongly asymmetric, with most hotspots concentrated at low intensities and a long tail of very high-intensity events — precisely the *outliers* that penalize centroid-based models. Geographically, hotspots form dense, contiguous patches in the Cerrado and the Amazon, interspersed with sparse occurrences, a pattern that motivates the use of density-sensitive algorithms.

B. Algorithm and Computational Environment Setup

The experiments were coded in Python 3.12.10, using the libraries *scikit-learn* 1.9.0, *pandas* 3.0.4, *numpy* 2.5.0, *scipy* 1.18.0, *matplotlib* 3.11.0, and *seaborn* 0.13.2. The execution environment was a notebook with an Intel Core i5-10210U processor (4 cores, 8 threads, base clock of 1.60 GHz and maximum of 2.10 GHz), 16 GB of RAM, running Microsoft Windows 10 Pro (build 10.0.19045).

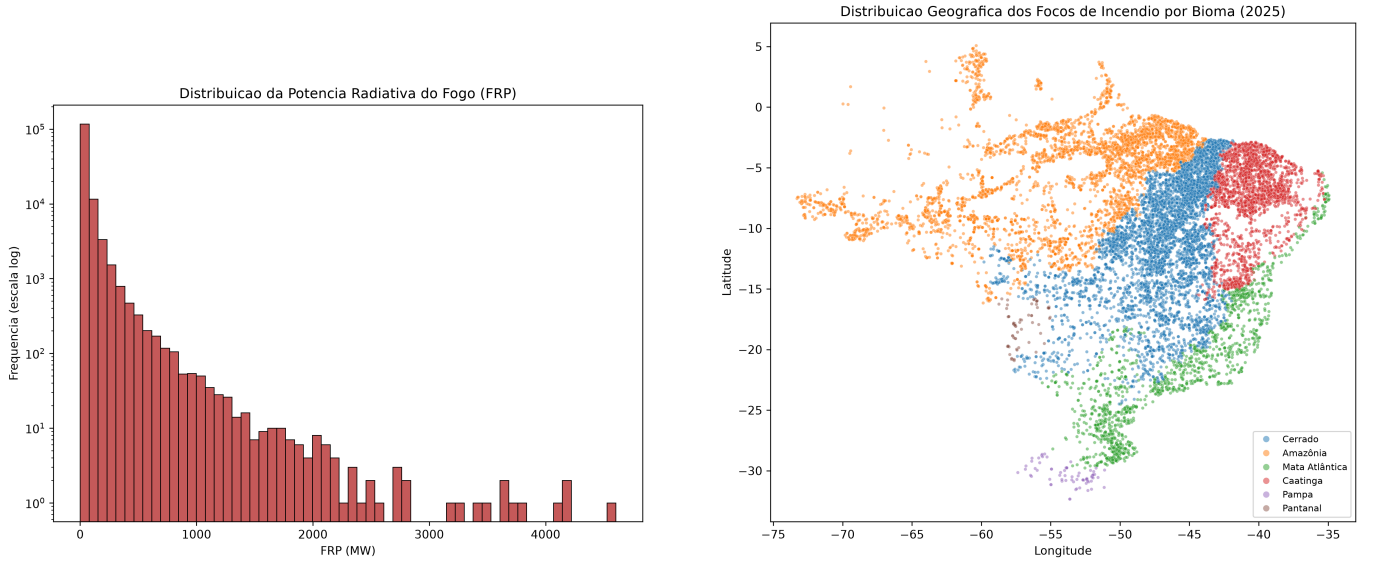


Figure 1. Exploratory data analysis. Left: FRP histogram (y -axis in logarithmic scale), showing strong asymmetry. Right: geographic distribution of hotspots colored by biome.

C. Analysis Criteria and Statistical Rigor

Ensuring the mathematical stability of results, the methods were tested in 30 independent iterations. In each round, 80% of the dataset was resampled using declared random seeds tied to an iterative variable, ensuring reproducibility. It is crucial to note that the *fit* of the algorithms occurred on this dense base of ≈ 109 thousand points, preserving the density structure required by DBSCAN. To mitigate the $\mathcal{O}(n^2)$ cost of the Silhouette Coefficient, metric computation used internal subsampling of 5,000 instances per cycle, while simultaneously monitoring execution time (in seconds). For DBSCAN, points labeled as noise (-1) were excluded from the silhouette calculation, since noise does not constitute a cohesive cluster; K-Means and GMM do not produce this label. Finally, the non-parametric Kruskal-Wallis test was applied to the silhouette distributions of the 30 runs, suitable because it does not assume data normality.

D. Results and Discussion

Table I summarizes the average performance over the 30 runs. The empirically collected data attested that centroid-based models prove insufficient for ecological phenomena of organic dispersion.

Table I
STATISTICAL SUMMARY OF THE 30 INDEPENDENT RUNS COMPARING SPATIAL CLUSTERING QUALITY AND COMPUTATIONAL COST.

Mean Metric (30x)	K-Means	GMM	DBSCAN
Silhouette Coef. (mean)	0.3554	0.0474	0.4619
Silhouette Coef. (median)	0.3553	0.0373	0.5177
Std. Dev. (Silhouette)	± 0.004	± 0.025	± 0.172
Mean time (s)	1.69	5.30	15.38
Std. Dev. (Time, s)	± 0.28	± 0.70	± 1.72

Traditional partitioning (K-Means) was hampered by the contorted fire patches and FRP outliers, achieving a silhouette

of 0.3554. Transitioning to probabilistic ellipses via GMM **degraded** the result (0.0474): the inclusion of the strongly asymmetric radiative intensity violates the Gaussian component assumption, dispersing probabilities. In notable superiority, DBSCAN achieved a mean silhouette of 0.4619 (median 0.5177), mapping continuous fire corridors and segregating isolated hotspots as noise (-1). In the adopted configuration, the algorithm identified 3 to 4 dense clusters, labeling only $\approx 1.4\%$ of instances as noise. The increased convergence time (15.38 s) reflects processing over the dense base, but does not preclude the solution for public managers. It should be noted that this mean value is lower than that recorded in the parameter variation study for the same $\epsilon = 0.30$ (0.6407): that corresponds to a single fixed-seed run, whereas the main experiment averages 30 independently resampled 80% subsets. The high standard deviation (± 0.172 , versus ± 0.004 for K-Means) highlights DBSCAN's sensitivity to sample composition, an instability that should be weighed in its operational adoption. Finally, the silhouette comparison is influenced by the number of clusters — DBSCAN forms 3 to 4 clusters, versus $k = 8$ for the others — which tends to favor configurations with fewer, well-separated groups; even so, its superiority holds even at the optimal k for K-Means.

The **Kruskal-Wallis hypothesis test** on the silhouettes of the 30 runs yielded a statistic $H = 57.32$ and p -value $= 3.6 \times 10^{-13}$. Since $p \ll 0.05$, the null hypothesis of equality among distributions is rejected, statistically proving that DBSCAN's advantage is not due to chance. Fig. 2 graphically illustrates this separation between the distributions.

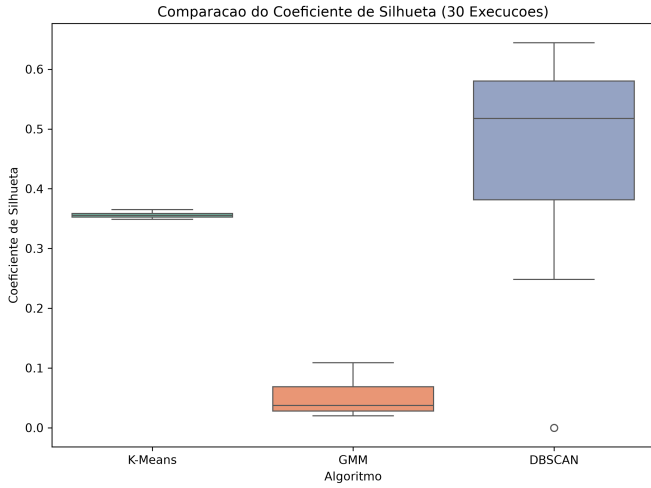


Figure 2. Distribution of the Silhouette Coefficient across 30 independent runs for each algorithm. DBSCAN concentrates at higher levels, with a notably higher median.

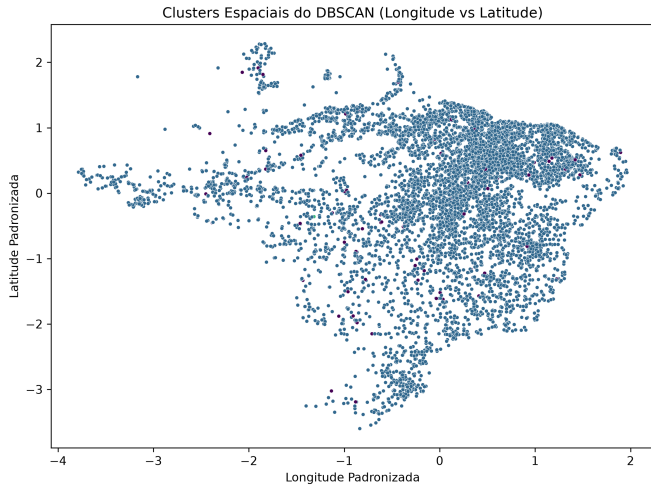


Figure 3. Geographic scatter of fire hotspots (standardized Longitude $\times$ Latitude), colored by clusters identified by DBSCAN. Continuous clusters and segregation of sparse hotspots are observed.

E. Parameter Variation Study

As required by the course, the influence of hyperparameters on clustering quality was evaluated (Fig. 4). For K-Means, the number of clusters $k \in [2, 12]$ was varied: the silhouette reaches its maximum at $k = 5$ (0.4049) and decays for more fragmented configurations. Note that even at its optimal point, K-Means remains well below DBSCAN, reinforcing that the limitation is not one of parameterization, but of topological assumption. For DBSCAN, the radius ϵ was varied: small values ($\epsilon = 0.10$) over-fragment the space and produce negative silhouette, while the metric grows monotonically up to the peak at $\epsilon = 0.30$ (0.6407), at which point only 1.3% of points are noise. This sensitivity analysis empirically justifies the adopted calibration and demonstrates the method's strong dependence on the choice of ϵ .

F. Segmented Analysis by Biome

To investigate how geography alters fire morphology, the analysis was repeated in isolation for the two most representative biomes. Table II reveals heterogeneous behavior: in the Cerrado, with more homogeneous vegetation, DBSCAN achieved a strong silhouette (0.5924); in the Amazon, where hotspot density is more diffuse, the DBSCAN index with the global configuration ($\epsilon = 0.30$) dropped to 0.2941, below K-Means. However, upon **recalibrating ϵ by biome**, the Amazon DBSCAN recovers to 0.4455 (with $\epsilon = 0.40$), again surpassing the local K-Means. This result demonstrates that the effectiveness of density models depends on the spatial continuity of each biome and that regionalized parameterization is a promising strategy.

Table II
SILHOUETTE COEFFICIENT BY BIOME. THE LAST COLUMN RECALIBRATES ϵ SPECIFICALLY FOR EACH BIOME.

Biome	K-Means	GMM	DBSCAN ($\epsilon = 0.30$)	DBSCAN (recal.)
Cerrado	0.3795	0.0260	0.5924	0.5924 $\epsilon = 0.30$
Amazon	0.3942	0.0829	0.2941	0.4455 $\epsilon = 0.40$

V. CONCLUSIONS

This work successfully consolidated the application of Artificial Intelligence toward the 2030 Agenda. The formulation and exhaustive testing on the INPE dataset empirically proved that extracting intelligence in ecosystems of high natural dispersion depends not on distance-based partitioning, but on organic coherence evaluated by density algorithms. DBSCAN, even with the feature vector extended by FRP, proved to be a justified, stable, and statistically superior instrument ($p < 10^{-12}$) for identifying active fronts of destruction. The parameter variation study confirmed that this advantage is structural — K-Means does not reach it even at its optimal k . The biome analysis, in turn, revealed that DBSCAN's superiority is sensitive to local geography, being expressive in the Cerrado and recoverable in the Amazon only through regionalized recalibration of ϵ .

As proposals for future work, the integration of the chronological registration variable available in the dataset is suggested, migrating the inference to a spatiotemporal context with the ST-DBSCAN algorithm [7], as well as automatic calibration of ϵ by biome, providing authorities with a mapping of not only space, but also temporal propagation speed.

REFERENCES

- [1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2011.
- [2] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. KDD*, 1996, pp. 226–231.
- [3] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.
- [4] National Institute for Space Research (INPE), "Queimadas Program / TerraBrasilis." [Online]. Available: <https://terrabrasilis.dpi.inpe.br/>

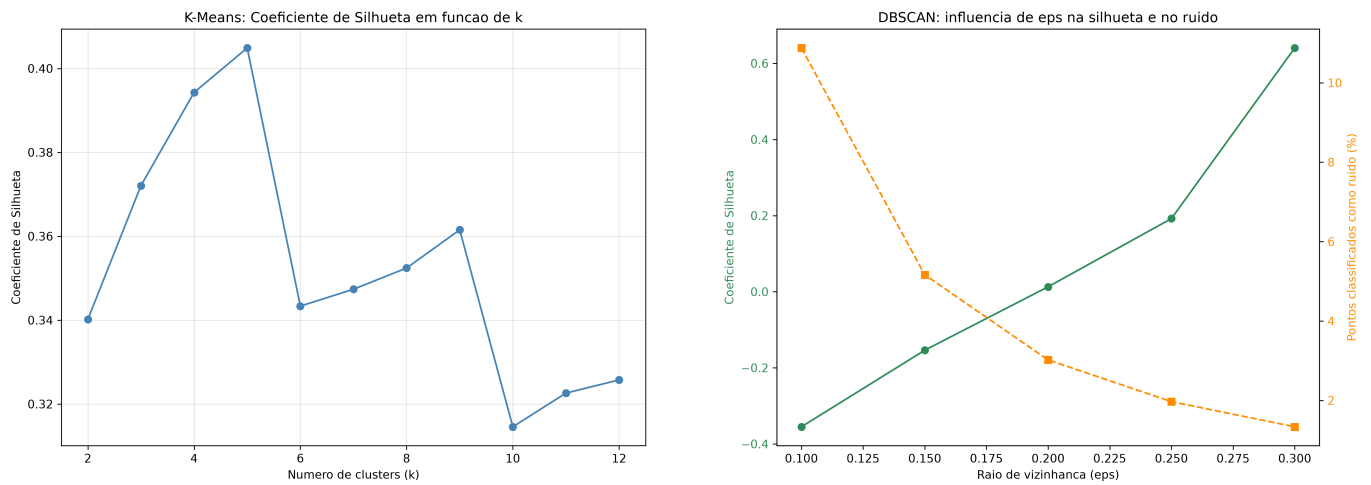


Figure 4. Parameter variation study. Left: K-Means silhouette as a function of k (peak at $k = 5$). Right: DBSCAN silhouette and noise percentage as a function of ϵ (peak at $\epsilon = 0.30$).

- [5] Esri Developer, "Detecting deforestation in the Amazon rainforest using unsupervised K-means clustering on satellite imagery," ArcGIS API for Python, 2024.
- [6] M. Ortega Adarme, R. Q. Feitosa, P. N. Happ, C. A. De Almeida, and A. R. Gomes, "Evaluation of deep learning techniques for deforestation detection in the Brazilian Amazon and Cerrado biomes from remote sensing imagery," *Remote Sensing*, vol. 12, no. 6, p. 910, 2020.
- [7] D. Birant and A. Kut, "ST-DBSCAN: An algorithm for clustering spatial-temporal data," *Data Knowl. Eng.*, vol. 60, no. 1, pp. 208–221, 2007.
- [8] N. Caron, H. Noura, C. Guyeux, and B. Aynes, "Localized forest fire risk prediction: A department-aware approach for operational decision support," arXiv preprint arXiv:2506.04254, 2025.
- [9] K. Faceli, A. C. Lorena, J. Gama, and A. C. P. L. F. de Carvalho, *Inteligência Artificial: Uma Abordagem de Aprendizado de Máquina*. Rio de Janeiro: LTC, 2011.